

Noether Sheet 1

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These were my solutions to the Noether Mentoring Scheme Sheet 1. Problem statements are abridged, and problem 1 is omitted because it isn't a math problem.

Problem 2

17:36 29/05/84 is a *pandigital* time and date as it uses each of the digits 0-9 exactly once.

What is the first pandigital time and date of each century?

Solution to problem 2

The day cannot be 30 or 31. Suppose the contrary, the tens digit of the hour is 0,1 or 2 and the tens digit of the month is 0 or 1. Thus the hour must use the number 2, so that the month can use 0 if the day is 31 and 1 if the day is 30. So now the numbers 0,1,2 and 3 are all taken. In the hour the only numbers that can follow that 2 are 0,1,2 and 3, but since all of them are taken we will repeat a digit, contradiction.

Now we notice that these positions can only contain numbers from the set $\{0, 1, 2\}$ (the tens digit of the month can't be 2 though).

$$\begin{array}{c} * \\ _ \end{array} : \begin{array}{c} * \\ _ \end{array} \begin{array}{c} * \\ _ \end{array} / \begin{array}{c} * \\ _ \end{array} \begin{array}{c} * \\ _ \end{array} / \begin{array}{c} * \\ _ \end{array}$$

Also if we make the tens digit of the hour a 2 then it must be followed by a 3 since 0,1 and 2 are taken, but if we make the tens digit of the hour a 1 then we can freely put whatever digit after it, allowing us to use the digit 3 in the tens digit of the year. This is optimal 3 is the smallest digit available to use. Finally if we make the tens digit of the hour a 0 then the month will be $\overline{1x} > \overline{0x}$ so this is unoptimal. Putting this information together we have

$$\begin{array}{c} 1 \\ _ \end{array} : \begin{array}{c} _ \\ _ \end{array} \begin{array}{c} 2 \\ _ \end{array} / \begin{array}{c} 0 \\ _ \end{array} \begin{array}{c} _ \\ _ \end{array}$$

Now we proceed greedily, it is optimal to make the year 34. We can't make the month 05 because otherwise the tens digit of the minutes would cause a repetition (the numbers 0,1,2,3,4 and 5 would be taken), so it is best to make the month 06. As reasoned the tens digit of the minutes must be 5 since there are no other options. The time and date now look like so

$$\begin{array}{c} 1 \\ _ \end{array} : \begin{array}{c} 5 \\ _ \end{array} \begin{array}{c} 2 \\ _ \end{array} / \begin{array}{c} 0 \\ _ \end{array} \begin{array}{c} 6 \\ _ \end{array} / \begin{array}{c} 3 \\ _ \end{array} \begin{array}{c} 4 \\ _ \end{array}$$

Now there are no more restrictions on the digits we can just greedily fill in the blanks and the final answer comes out as

$$\begin{array}{c} 1 \\ _ \end{array} \begin{array}{c} 8 \\ _ \end{array} : \begin{array}{c} 5 \\ _ \end{array} \begin{array}{c} 9 \\ _ \end{array} \begin{array}{c} 2 \\ _ \end{array} \begin{array}{c} 7 \\ _ \end{array} / \begin{array}{c} 0 \\ _ \end{array} \begin{array}{c} 6 \\ _ \end{array} / \begin{array}{c} 3 \\ _ \end{array} \begin{array}{c} 4 \\ _ \end{array}$$

■

Comments: This problem looks deceptively easy. I fakesolved many times, I wonder if there is a more elegant approach here.

Problem 3

Derive the quadratic formula for $ax^2 + bx + c$ given $b^2 \geq 4ac$.

Solution to problem 3

Complete the square,

$$\begin{aligned}a \left(x + \frac{b}{2a} \right)^2 &= \frac{b^2 - 4ac}{4a} \\ \sqrt{a} \left(x + \frac{b}{2a} \right) &= \pm \frac{1}{2\sqrt{a}} \sqrt{b^2 - 4ac} \\ x &= -\frac{b}{2a} \pm \frac{1}{2a} \sqrt{b^2 - 4ac}\end{aligned}$$

as required. ■

Comments: This is a pretty standard problem and I've seen the derivation before so it was solved quickly.

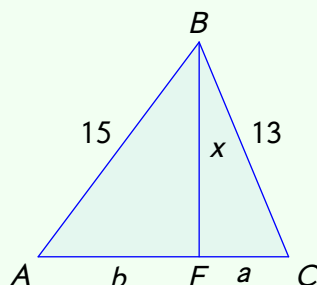
Problem 4

A triangle has sides of length 13 cm, 14 cm and 15 cm.

- Show that this triangle can be split into two right angled triangles with integer side lengths (when measured in centimetres).
- A rectangle is drawn inside the triangle, with its base on one of the sides of the triangle. What is the largest possible area of this rectangle?

Solution to problem 4

(a): Let $\triangle ABC$ be a triangle such that $AB = 15$ cm, $CA = 14$ cm and $BC = 13$ cm. Let F be the foot of the altitude that passes through B . Let $FC = a$ cm and $FA = b$ cm, and notice that $\triangle BFC$ and $\triangle BFA$ are both right angled triangles. Let $BF = x$ cm.



We have a system of equations.

$$a^2 + x^2 = 13^2$$

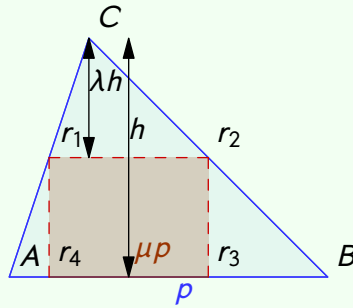
$$b^2 + x^2 = 15^2$$

and so $b^2 - a^2 = (b - a)(14) = 56$ since $b + a = 14$. So we deduce that $b - a = 4$, from which we can solve to see that $b = 9$, $a = 5$ and also $x = \sqrt{13^2 - 5^2} = 12$. Thus the triangles $\triangle BFC$ and $\triangle BFA$ are both right-angled and also have integer side lengths. ■

(b): We begin with a claim.

Claim 1: For any non-degenerate triangle $\triangle ABC$, let R be a rectangle inscribed within $\triangle ABC$ such that one of its bases lies on one of the sides of ABC . Then $\text{Area}(R) \leq \frac{1}{2} \text{Area}(\triangle ABC)$.

Proof: Let the base of R lie on some side P of $\triangle ABC$ that has length p . Let the height of $\triangle ABC$ be h (modelling P to be the base of the triangle). Let R have vertices r_1, r_2, r_3 and r_4 , (starting from the top left and going clockwise). Let the height of R be $(1 - \lambda)h$ and let the length of R be μp for some $0 < \mu, \lambda < 1$. The following diagram can help us visualise,



Using corresponding angles and the fact that $\angle ACB$ is common to both triangles we deduce that $\triangle ABC$ and $\triangle Cr_1r_2$ are similar, thus it is the case that $\lambda h/h = \mu p/p$. Thus it is the case that $\mu = \lambda$. The area of R is $\mu p \cdot (1 - \lambda)h = \mu(1 - \mu)ph$. It is easy to check that the maxima of $f(x) = x(1 - x)$ is $\frac{1}{4}$ and so we find that

$$\text{Area}(R) \leq \frac{1}{4}ph = \frac{1}{2}\text{Area}(\triangle ABC). \quad \square$$

Now due to part (a) we know that $\text{Area}(\triangle ABC) = \frac{1}{2}(12)(14) = 84$, and so by claim 1 the largest possible area of the rectangle is 42. ■

Comments: I think this problem was my favourite just after problem 8. The motivation for conjecturing $\text{Area}(R) \leq \frac{1}{2}\text{Area}(\triangle ABC)$, came from considering a rectangle with base on the side AC using the triangle in part (a). One can form a quadratic for the area of the rectangle in terms of the height of the rectangle which attains a maxima of 42. Now that answer is fairly suspicious since $42 = \frac{1}{2}84 = \frac{1}{2}\text{Area}(\triangle ABC)$, so from here it is natural to guess that claim 1 might be true.

Problem 5

- (a) Aadya cycles down a hill at x m/s and up the same hill at y m/s. Barry then estimates her average speed by calculating $\frac{x+y}{2}$ m/s. Show that Barry's estimation is always greater than or equal to Aadya's true average speed.
- (b) One day, Aadya cycled down the hill at x m/s and up the hill at 3 m/s. Both x and the difference between Barry's estimation of Aadya's average speed and her true average speed are integers. What are the possible values of x ?

Solution to problem 5

(a): Suppose that the hill is h metres long. For some real positive times t_1, t_2 we have $x = h/t_1$, $y = h/t_2$. The average speed of Aadya is

$$\bar{s} := \frac{2h}{t_1 + t_2} = \frac{2xy}{x + y}$$

Now consider

$$\begin{aligned} \frac{2xy}{x + y} &\leq \frac{x + y}{2} \\ 4xy &\leq (x + y)^2 \\ 2xy &\leq x^2 + y^2 \end{aligned}$$

but this is true by AM-GM. ■

(b): Similarly we write $x = h/t_1$, $3 = h/t_2$. The average speed of Aadya is

$$\bar{s} := \frac{2h}{t_1 + t_2} = \frac{6x}{x + 3}$$

Now we are told that

$$\frac{x + 3}{2} - \frac{6x}{x + 3} = \frac{(x - 3)^2}{2(x + 3)} \in \mathbb{Z}_{\geq 0}$$

since Barry's estimation is always greater than or equal to Aadya's true speed. We are asking to find all $x \in \mathbb{Z}_{\geq 0}$ such that $2(x + 3) \mid (x - 3)^2$. If x were even then the numerator isn't divisible by 2 whilst the denominator is divisible by 2, a contradiction. So x must be odd. Observe that

$$\frac{(x - 3)^2}{2(x + 3)} = \frac{x - 9}{2} + \frac{36}{2x + 6}.$$

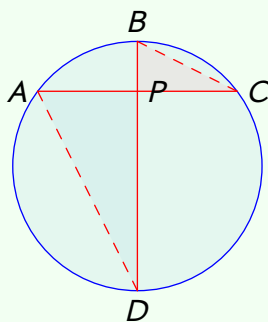
Since x is odd the first term is an integer so we just require $2x + 6$ to be a factor of 36. Thus $2x + 6 \in \{1, 2, 3, 4, 6, 9, 12, 18, 36\}$. Since $2x + 6$ is even we can eliminate any odd candidates, and since $x \geq 0$ we have $2x + 6 \geq 6$. Thus $2x + 6 \in \{6, 12, 18, 36\}$. So we solve to see that $x = 0, 3, 6, 15$. ■

Problem 6

Prove the Power of a point theorem.

Solution to problem 6

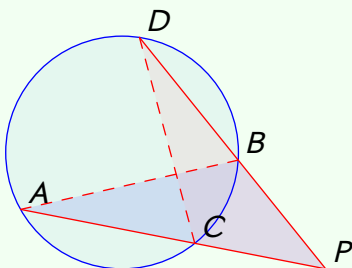
(a): Consider constructing $\overline{DA}$ and $\overline{BC}$,



Due to the Inscribed angle theorem $\angle ADP = \angle BCP$, and since opposite angles are equal $\angle APD = \angle BPC$. Since there are two equal angles and angles in a triangle add to 180° we also have $\angle DAP = \angle PBC$. Now since there are three equal angles the triangles $\triangle DAP$ and $\triangle PBC$ are similar.

Thus $AP/DP = BP/CP$ which re-arranges to $AP \cdot CP = BP \cdot DP$.

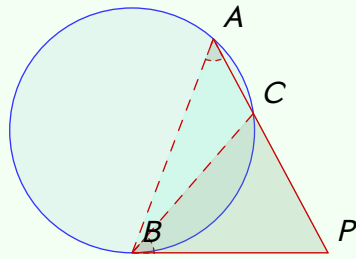
(b): Consider constructing the triangles $\triangle ABP$ and $\triangle DCP$.



Again due to the Inscribed angle theorem $\angle PAB = \angle CDP$. Since BP and CP are common to both triangles we deduce that $\angle APB = \angle DPC$. Since there are two equal angles and angles in a triangle add up to 180° we also have $\angle ABP = \angle PCD$. Now since there are three equal angles the triangles $\triangle ABP$ and $\triangle DCP$ are similar.

Thus $AP/BP = DP/CP$ which re-arranges to $AP \cdot CP = BP \cdot DP$.

(c): Consider constructing the triangles $\triangle ABP$ and $\triangle BCP$.



Now clearly $\angle BPC = \angle APB$ because BP and CP are common to both triangles. Then also $\angle PBC = \angle BAP$ due to the Alternate segment theorem. Since there are two equal angles and angles in a triangle add up to 180° we also have $\angle ABP = \angle BCP$. Since we have three equal angles triangles $\triangle ABP$ and $\triangle BCP$ are similar.

Thus $AP/BP = BP/CP$ which re-arranges to $AP \cdot CP = BP^2$. ■

Problem 7

Consider the thirteen factorials $0!, 1!, 2!, \dots, 12!$. We want to divide these up into three sets of size 4 and one set of size 1. Moreover, we would like the product of the elements in each of the sets of size 4 to be a square number.

In how many ways can this be achieved?

Solution to problem 7

The following table will be of use to us, even numbers are in red whilst odd numbers are in blue.

n	0	1	2	3	4	5	6	7	8	9	10	11	12
$v_2(n!)$	0	0	1	1	3	3	4	4	7	7	8	8	10
$v_3(n!)$	0	0	0	1	1	1	2	2	2	4	4	4	5
$v_5(n!)$	0	0	0	0	0	1	1	1	1	1	2	2	2
$v_7(n!)$	0	0	0	0	0	0	0	1	1	1	1	1	1
$v_{11}(n!)$	0	0	0	0	0	0	0	0	0	0	0	1	1

We define the *score* of a set, to be the product of the elements in that set.

Claim 1: The number in the set of size 1 must be $6!$.

Proof: Suppose the three sets of size 4 have scores A^2, B^2 and C^2 . Then suppose the number we place in the set of size 1 is $k!$. We observe that $A^2 B^2 C^2 k! = 0! \cdot 1! \cdot 2! \cdots 12!$. Let $N = (0! \cdot 1! \cdot 2! \cdots 12!)/k! = (ABC)^2$. Since $v_5(0! \cdot 1! \cdot 2! \cdots 12!) = 11$, $k!$ must have an odd 5-adic valuation which means $k \in \{5, 6, 7, 8, 9\}$,

If $k = 7, 8, 9$ then $v_7(N) = 5$ so N isn't square. If $k = 6$ then N is in fact a square. If $k = 5$ then $v_3(N) = 25$ so N isn't square. Thus $k = 6$. $\square$

Claim 2: Let p be some prime. There must be an even number of factorials that have an odd p -adic valuation in each set of size 4, similarly there must be an even number of factorials that have an even p -adic valuation in each set of size 4.

Proof: Suppose there are x factorials that have an even p -adic valuation and y factorials that have an odd p -adic valuation in one set of size 4. Let S be the score of that set. Then $v_p(S) \equiv y \equiv 0 \pmod{2}$ since perfect squares only have even p -adic valuations. The claim follows from the fact that y is even. $\square$

Claim 3: $11!$ and $12!$ must be in the same set.

Proof: Since $11!$ and $12!$ are the only numbers with factors of 11, and they both have exactly one factor of 11, they must be in the same set otherwise one of the sets would have a score with an odd 11-adic valuation, but since the scores are all perfect squares this is a contradiction. $\square$

Claim 4: The set that contains $11!$ and $12!$ is either $\{2!, 3!, 11!, 12!\}$ or $\{2!, 4!, 11!, 12!\}$.

Proof: Now in the set that contains $11!, 12!$ the two other numbers $x!$ and $y!$ must satisfy $v_3(x! \cdot y!) \equiv 1 \pmod{2}$ because $v_3(11! \cdot 12!) = 9$ is odd and $x! \cdot y! \cdot 11! \cdot 12!$ is a perfect square. Call a number 3-*blue* if it has an odd 3-adic valuation and is not one of $11!, 12!$. Call a number 3-*red* if it has an even 3-adic valuation and is not one of $11!, 12!$.

Out of $x!$ and $y!$ exactly one of them must be 3-blue because otherwise $x! \cdot y!$ would have an even 3-adic valuation. The only 3-blue numbers are $3!, 4!$ and $5!$ and they all have an odd 2-adic valuation. Suppose WLOG that $x!$ is 3-blue. Then $y!$ must have an odd 2-adic valuation also because $v_2(11! \cdot 12! \cdot x!)$ is odd and $x! \cdot y! \cdot 11! \cdot 12!$ is a perfect square. We can use our table and the information deduced thus far to see that $y! \in \{2!, 8!, 9!\}$. However $y! \neq 8!, 9!$ because then $v_5(x! \cdot y! \cdot 11! \cdot 12!) = 3$ which is odd, but this is a contradiction. Thus $y! = 2!$. Now it can be checked that the set that contains $11!, 12!$ must be either $\{2!, 3!, 11!, 12!\}$ or $\{2!, 4!, 11!, 12!\}$. $\square$

Now suppose that we have placed $6!$ in the set of size 1, and we have selected the numbers that will be in the set that contains $11!$ and $12!$ using claim 4. The only remaining numbers that have factors of 7 are $7!, 8!, 9!$ and $10!$. Let's call these numbers *elegant*. Notice that elegant numbers all have exactly one factor of 7. Now due to claim 2 each set must contain an even number of elegant numbers. It is important to note that we can't have all four elegant numbers in one set since $v_3(7! \cdot 8! \cdot 9! \cdot 10!) = 5$ is odd and so the product of the four elegant numbers isn't a perfect square.

One of the sets will look like this $\{10!, x!, _, _\}$ where $x!$ is an elegant number other than $10!$. Since $v_5(10! \cdot x!) = 3$, $5!$ must be in this set because it is the only remaining number with a factor of 5. Thus that set looks like $\{10!, x!, 5!, _\}$. However now using the table it can be checked that $v_3(10! \cdot x! \cdot 5!)$ is odd and so the final number represented by $_$ must have an odd 3-adic valuation. That means it must be $3!$ or $4!$, whichever one wasn't used in the set that contains $11!$ and $12!$.

That means the third and final set must look like $\{0!, 1!, y!, z!\}$ where $y!$ and $z!$ are the two remaining unused elegant numbers. One can use the table to check that only $(y!, z!) = (8!, 9!)$ works. This means that $x! = 7!$.

Now putting all of this together our sets look like so, let $a!, b!$ be $3!, 4!$ in some order, $\{6!\}$, $\{0!, 1!, 8!, 9!\}$, $\{2!, a!, 11!, 12!\}$ and $\{b!, 5!, 7!, 10!\}$. Now it can be checked that, no matter the order of $a!, b!$, each set has a score that is a perfect square. There are 2 ways to choose $(a!, b!)$ so the final answer is 2. $\blacksquare$

Comments: This problem is arguably the hardest on the sheet, the solution I found involves some facts from number theory and a lot of patience to work through all of the cases.

Problem 8

Find all sets of three consecutive positive integers such that each is either a prime or a power of a prime. (We do not regard 1 as a power of a prime in this question.)

Solution to problem 8

Call a set *good* if it contains three consecutive positive integers such that they are all prime or powers of a prime. Let $n \in \mathbb{N}$ such that $n, n+1$ and $n+2$ are all primes or powers of a prime.

If n is even then $n = 2$ since 2 is the only even prime. Also since $n+1 = 3$ and $n+2 = 2^2$ the set $\{2, 3, 4\}$ is a good set.

If n is odd then $n+1$ is even which means it must be of the form 2^k for $k \in \mathbb{N}$. Similarly $n = 2^k - 1$ must be of the form p^a and $n+2 = 2^k + 1$ must be of the form q^b for odd primes p, q and $a, b \in \mathbb{N}$. Now we notice that $n = 2^k - 1 \equiv (-1)^k - 1 \pmod{3}$ and $n+2 = 2^k + 1 \equiv (-1)^k + 1 \pmod{3}$. Thus if k is even then $3 \mid n$, and if k is odd then $3 \mid (n+2)$.

Suppose that k is even, then we have $n = 3^a$ and $n+1 = 2^k$. We deduce that $2^k - 3^a = 1$ or equivalently $(2^{k/2} - 1)(2^{k/2} + 1) = 3^a = n$. Now we have two powers of 3 that are 2 apart, this only happens for $(3^0, 3^1)$ because powers of 3 grow too quickly past this point. So we have $2^{k/2} - 1 = 1$ and $2^{k/2} + 1 = 3$ which gives $a = 1, k = 2$ and so $n = 3$. Also since $n+1 = 2^2$ and $n+2 = 5$ the set will be good. So the only good set in this case is $\{3, 4, 5\}$.

Suppose that k is odd, then $n+2 = 3^b$ and $n+1 = 2^k$. This time we have $3^b - 2^k = 1$. We use LTE to finish. We need that $v_2(3^b - 1^b) = k$, there are two cases. If b is odd then $v_2(3^b - 1^b) = v_2(3 - 1) = 1 = k$. We find that $k = 1, b = 1$ but that implies that $n = 1$ which is not a power of a prime in the context of this question.

Thus b must be even and we have $v_2(3^b - 1^b) = v_2(3 - 1) + v_2(3 + 1) + v_2(b) - 1 = k$ which re-writes as $v_2(b) = k - 2$. So we find that $2^{k-2} \mid b$ and so $b \geq 2^{k-2}$. Clearly this inequality will be very restrictive since

$$1 = 3^b - 2^k \geq 3^{2^{k-2}} - 2^k$$

will only hold for $k \leq 3$ since that RHS grows too quickly. We have already eliminated the case $k = 1$ and it can be checked that $k = 3, b = 2$ is a valid solution here. Thus $n = 7$, and since $n+1 = 2^3$ and $n+2 = 3^2$ the set $\{7, 8, 9\}$ is the only good set here.

Putting all of that together the only good sets are $\{2, 3, 4\}$, $\{3, 4, 5\}$ and $\{7, 8, 9\}$. ■

Comments: This one was fun! The case when k is odd is probably the most challenging part of the problem, I personally can never remember all of the edge cases of LTE so I had to look it up. Of course my arguments that say something like "...grows too quickly" are not at all rigorous but I omit a formal justification because it'd make the solution really long, but I don't mind adding justification if you want!